

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1767

Gorge 18 138 kV
July 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1767. The Interconnection Customer proposes a 140.2 MW battery energy storage facility interconnecting at the Gorge 18 138 kV substation in the Pennsylvania area. The request was received on June 04, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 589 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,578,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gorge 18 – Byberry 138 KV	Temple – Sahtlam 138 KV	112.2	191	100.0
Byberry – Neshaminy 138 KV	Sahtlam Substation – Temple 138 KV	111.8	230	57.1
Felida – W Gallatin 345 KV	Wallenpaupack 500 kV/345 kV Auto	100.3	1,858	8.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gorge 18 – Byberry 138 KV	Rebuild line with higher-capacity conductor	\$1,422,000	100% (\$1,422,000)
Byberry – Neshaminy 138 KV	Rebuild line with higher-capacity conductor	\$4,874,000	100% (\$4,874,000)
Felida – W Gallatin 345 KV	Reconductor line	\$18,275,000	2% (\$282,000)
	Total allocated to project		\$6,578,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.